When adding UNIQUE or NOT NULL constraints to a table, all rows are checked immediately and the check process cannot be configured only for new modifications like new inserts, updates, or deletes.

On a large and busy database this can make it difficult to add UNIQUE and NOT NULL constraints since it might take a long time to check all of the rows. While all of the rows are being checked, the table is locked and blocking other writers and readers.

Those blocked queries will start to show up as application errors and cause user experience problems and possibly data loss. Thus adding NOT NULL and UNIQUE constraints to a large running database is considered a potentially unsafe operation. Is there a way to work around this? Check constraints to the rescue!

Instead of directly adding NOT NULL and UNIQUE constraints, first you'll add a CHECK constraint and mark it NOT VALID. Note that Foreign Key constraints also support the NOT VALID option.

To practice adding Check constraints, you'll add some to Rideshare. The examples were inspired by *Postgres Constraints for Newbies* from Crunchy Data.[5]

You'll add Check constraints to the Vehicle Reservations table.

To get started, take a look at the Check constraints on the trips table.

A Trip is completed and completed_at is set. This time value should always be *after* the time the trip was created, which is stored in the created_at column. This is enforced with a Check constraint in Rideshare.

Review the Migration to see the Check constraint definition, or review the equivalent SQL below.

Rails 6.1 natively supports Check constraints in Active Record sytax.[6] Support for if_exists[7] was added later.

sql/check_constraint_trips.sql

```
ALTER TABLE trips
ADD CONSTRAINT trips_completed_after_created
CHECK (completed_at > created_at);
```

An equivalent Active Record Migration would look like this:

5. https://www.crunchydata.com/blog/postgres-constraints-for-newbies
6. https://blog.saeloun.com/2021/01/08/rails-6-check-constraints-database-migrations
7. https://github.com/rails/rails/commit/25f97a66bdae6efe788b2d0ab7ab9cef6fc5a23a